Q1. Linux file operations

Original question: Create a directory named EHF_Practical and a file named evidence.txt. Write and display its contents, create a copy, rename the copied file, and verify the final directory contents using appropriate Linux commands.

Aim

Create EHF_Practical and evidence.txt; write and display content, copy it, rename the copy, and verify the final files.

Requirements

Kali/Linux, Bash, an ordinary user account and a writable home directory. No elevated privilege is required. Use a fresh directory; if EHF_Practical already exists, inspect it before continuing.

Procedure

Step 1 - Create and enter the directory

```
cd "$HOME"
mkdir EHF_Practical
cd EHF_Practical
pwd
```

`mkdir` creates the directory; `cd` changes the current directory; `pwd` confirms the full location. Stop if creation failed because the directory already exists: do not overwrite earlier work blindly.

Step 2 - Write and read the file

```
printf '%s\n' 'EHF laboratory evidence sample.' > evidence.txt
cat evidence.txt
```

`printf` writes one newline-terminated line. `>` creates the file or replaces its contents. `cat` displays the bytes as text. To add another line without replacing earlier content, use `>>` instead.

Step 3 - Copy, rename and verify

```
cp -i evidence.txt evidence_copy.txt
mv -i evidence_copy.txt evidence_backup.txt
ls -l
cmp evidence.txt evidence_backup.txt
printf 'Comparison exit status: %s\n' "$?"
```

`cp` creates a second file; `mv` changes the copy's name. `-i` prompts before an existing destination is overwritten. `ls -l` lists names, permissions, owners and sizes. `cmp` produces no text when the two files match; its immediate exit status is 0 for a match, 1 for a difference and greater than 1 for an error.

Expected output - illustrative

```
EHF laboratory evidence sample.
# Final names shown by ls:
evidence.txt
evidence_backup.txt
Comparison exit status: 0
```

The listing above abbreviates `ls -l`; timestamps and owners depend on your machine. evidence_copy.txt should be absent after the rename.

Observation and result

Record the actual path and final names. Both files should contain the same text and have equal byte sizes. The copy was renamed without changing the original. This completes the requested create, write, display, copy, rename and verification operations.

Common errors

- Permission denied: work in your home directory and inspect permissions; sudo is unnecessary.
- No such file: use pwd and ls to confirm your current directory and spelling.
- Existing destination: answer the overwrite prompt carefully or choose a fresh workspace.

Viva questions

- What is the difference between cp and mv? cp creates a copy; mv moves or renames a directory entry.
- What do > and >> do? They replace and append output, respectively.
- Why use cmp? It verifies file contents byte-for-byte rather than relying on names or sizes.
- Does ls -l show file contents? No; it shows directory entries and file attributes.